Deauthentication Cascades as a Global Security Risk: A Theoretical Framework for Individual-to-Planetary Authentication Collapse Events

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Keywords: continuous authentication, deauthentication cascade, cyber-physical systems, environmental interference, identity substrate, epistemic collapse, resilience engineering

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Abstract
As societies move toward ubiquitous continuous-authentication ecosystems — including bio-integrated wearables, ambient computing, and identity-linked automation — the stability of authentication signals becomes a critical dependency for civilizational function. This paper explores the theoretical possibility of deauthentication attacks, ranging from localized single-user disruptions to planet-scale authentication collapse events. These attacks may arise from hostile interference, unknown technologies, or unforeseen environmental anomalies. We analyze the systemic consequences of sudden identity-signal degradation, model the cascading failures that follow, and propose resilience frameworks to mitigate catastrophic outcomes.

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1. Introduction: Authentication as a Civilizational Substrate
Modern cybersecurity treats authentication as a discrete event. Emerging architectures instead treat it as a continuous physiological and behavioral signal, binding identity to:

- biometric telemetry
- cognitive signatures
- gait and kinetic patterns
- environmental co-presence graphs

As these systems proliferate, authentication becomes a background assumption of daily life. Devices, infrastructure, and services operate under the premise that identity is always known.

This creates a new class of systemic risk:
> What happens when identity itself becomes unreadable?

A sudden, widespread loss of authentication — whether caused by hostile action, environmental anomaly, or unknown technological interference — becomes a societal-scale failure mode.

This paper formalizes that risk.

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2. Defining the Deauthentication Attack

A deauthentication attack is any event in which the authentication substrate of a person or population becomes unreadable to the systems that rely on it.

We define three tiers:

Tier I — Localized Individual Deauthentication

A single user’s identity signal collapses.
Possible causes: sensor malfunction, localized interference, physiological anomaly.

Tier II — Regional or Population-Scale Deauthentication

A neighborhood, city, or nation experiences simultaneous authentication failure.
Possible causes: directed interference, environmental anomaly, systemic software fault.

Tier III — Planetary Authentication Collapse

All continuous-authentication systems simultaneously lose the ability to resolve identity.
Possible causes: unknown technologies, global environmental events, catastrophic systemic bug.

Each tier produces qualitatively different failure cascades, but all share a common structure:
identity loss → misclassification → defensive escalation → societal disruption.

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3. The Identity-Signal Collapse Mechanism

Continuous authentication systems extract a high-dimensional identity vector:

$$
S_{user}(t) = f(\text{biometric}, \text{behavioral}, \text{cognitive}, \text{environmental})
$$

A deauthentication event is characterized by a rapid degradation:

$$
\Delta S = \left| \frac{d}{dt} S_{user}(t) \right|
$$

If the rate of collapse exceeds the system’s tolerance threshold τ_{crit}, the system interprets the event as:

- identity theft
- physical replacement
- hostile intrusion

This is the False Invader Condition.

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4. The Deauthentication Cascade
Once the system enters the False Invader Condition, a predictable cascade unfolds.

4.1 Zero-Trust Severance
All privileges are revoked.
The system assumes the user is no longer the legitimate operator.

4.2 Environmental Lockout
Smart infrastructure denies access:

- transit
- doors
- communication systems
- medical devices
- financial systems

The user becomes a non-person in the eyes of the infrastructure.

4.3 Defensive Escalation
High-security systems may escalate to defensive postures.
This paper treats such escalation abstractly, without operational detail.

4.4 Epistemic Contagion
Humans observing machine misclassification may adopt the same belief:

> “If the system says you’re not you, maybe you’re not.”

This produces machine-induced paranoia.

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5. Planetary-Scale Authentication Collapse
A global deauthentication event — whether caused by unknown physics, hostile interference, or emergent systemic failure — produces a unique class of civilizational risk.

5.1 Infrastructure Paralysis
Every system that requires identity halts simultaneously:

- transportation
- logistics

- healthcare
- energy
- communication
- governance

5.2 Global Epistemic Crisis
If all devices misclassify all humans, the world experiences:

- universal lockout
- universal misidentification
- universal defensive escalation

This is a planetary epistemic inversion:
machines lose the ability to distinguish humans from threats.

5.3 Sociotechnical Shockwave
The psychological impact is immediate:

- trust collapse
- communication breakdown
- mass confusion
- spontaneous social fragmentation

This is not merely a cybersecurity failure — it is a civilizational stress event.

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6. Unknown or Conceptually Conceivable Technologies
This paper does not assert the existence of any specific technology.
Instead, it explores conceptually conceivable mechanisms that could cause authentication collapse:

- wide-area signal dampening
- environmental noise spikes
- exotic interference phenomena
- systemic software faults
- emergent behavior in distributed AI systems
- unknown physical interactions with biometric sensors

These mechanisms are treated as theoretical stressors, not operational tools.

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7. Resilience and Mitigation Frameworks

To prevent catastrophic outcomes, future systems must incorporate doubt, redundancy, and contextual awareness.

7.1 Environmental Canary Arrays
Detect environmental anomalies before identity logic collapses.

7.2 Multi-Modal Identity Anchors
Use orthogonal identity channels that cannot all fail simultaneously.

7.3 Mesh Consensus Authentication
Require quorum-based agreement before misclassification.

7.4 Aggression-Inhibition Logic
Ensure systems default to safe mode, not escalation, when identity is uncertain.

7.5 Temporal Smoothing of Identity Signals
Prevent instantaneous drops from triggering threat logic.

7.6 Human-Override Windows
Allow humans to assert identity during anomalous conditions.

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8. Discussion: Why This Threat Class Matters
Even if the probability of a planetary deauthentication event is extremely low, the impact is extremely high.
This places the scenario in the category of:

- high-impact, low-probability risks
- systemic fragility analysis
- resilience engineering
- civilizational safety research

The purpose of this paper is not to predict such events, but to ensure that if they occur, society is not unprepared.

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9. Conclusion
As authentication becomes continuous, embodied, and ubiquitous, the stability of identity signals becomes a foundational requirement for societal function. Deauthentication attacks — whether localized or planetary — represent a class of systemic risks that must be studied, modeled, and mitigated. The future of cyber-physical security depends on designing systems

that treat identity uncertainty not as a trigger for escalation, but as a signal to enter safe, reversible, human-centered modes of operation.